

Chapter 5

Estimation of Seasonal Variation

Scope of Study

5.1 Estimation of Seasonal Variation

5.2 Estimation of Variation C and I

5.1 Estimation of Seasonal Variation

Seasonal variation refers to changes in data that occur due to seasonal influences and repeat at regular intervals. The seasonal period is typically shorter than one year. These variations can take different forms. In this study, we will consider two types: seasonal variation that remains relatively constant or changes only slightly over time, called "**additive seasonal variation**," and time series in which seasonal variation is proportional to the level of the series, called "**multiplicative seasonal variation**." The magnitude of this variation is commonly measured using an index called the "**seasonal index**." Examples of seasonal variation are shown in Figures 1 through 4.

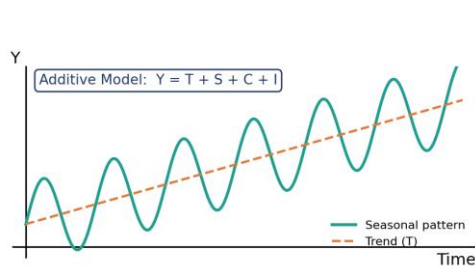


Figure 1 shows the form of Additive Seasonal Variation

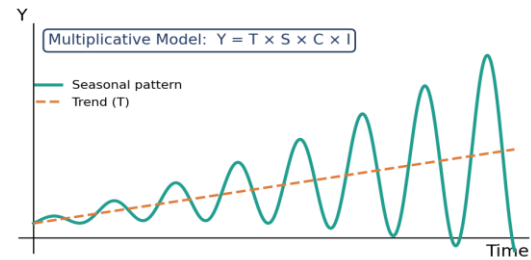


Figure 2 shows the form of Multiplicative Seasonal Variation

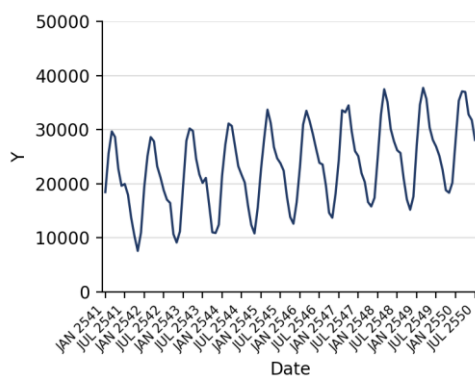


Figure 3 Shows the seasonal variation in the volume of shrimp exports from January 2541 – December 2550

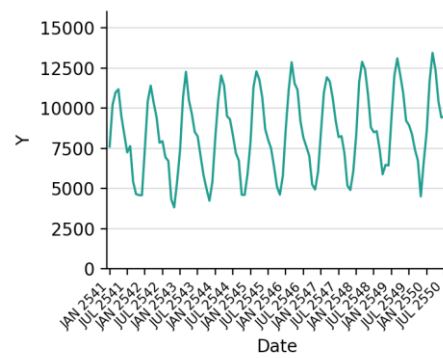


Figure 4 Shows the seasonal variation in the value of shrimp exports from January 2541 – December 2550

In measuring seasonal variation, the "**seasonal index**" is commonly used. This index represents the percentage of influence attributable to seasonal variation relative to the average of the data.

- If the seasonal index for a particular month (or quarter) equals **104%**, this means that the effect of seasonal variation causes the data for that month (or quarter) to be 4% higher than normal ($104 - 100 = 4\%$).
- If the seasonal index for a particular month (or quarter) equals **93%**, this means that the effect of seasonal variation causes the data for that month (or quarter) to be 7% lower than normal ($100 - 93 = 7\%$).
- There are two methods for calculating the seasonal index:
 1. Ratio-to-Moving Average Method
 2. Ratio-to-Trend Method

● **Estimation of Multiplicative Seasonal Variation Using the Moving Average Method**

The procedure can be summarized as follows:

- 1) Isolate the trend component (T) from the original data **using the moving average method**

Suppose that

The number of periods in the moving average = the length of one complete seasonal cycle.

- If the annual data **consists of 4 periods** (4 quarters per year), use a **4-period moving average**.
- If the annual data **consists of 12 periods** (12 months per year), use a **12-period moving average**.

Note: If the number of **periods** in the moving average is even, a **centered moving average** must be calculated.

- 2) **Divide the original data by the corresponding moving average values** (Y) and multiply by 100

- 3) Calculate the average of the results from Step 2

- 4) Verify that **the sum of the period averages** from Step 3 equals the percentage corresponding to the number of periods in one seasonal cycle

- If **the sum of averages** equals 400 (for quarterly data) or 1200 (for monthly data), the average for each period represents the "**multiplicative seasonal index**".
- If **the sum of averages does not equal** 400 (for quarterly data) or 1200 (for monthly data), the averages for each period **must be adjusted**. The adjusted values will then represent the desired **multiplicative seasonal index**.

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Example The sales manager of a department store has collected quarterly sales data (every 3 months) from B.E. 2537–2542. Demonstrate how to calculate the multiplicative seasonal index using the ratio-to-moving average method.

Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$\frac{Y}{CMA} \times 100$
2537	1	6.7			-	-
	2	4.6			-	-
	3	10.0	34.0	$(34.0 / 4)$ $= 8.500$	8.4750	117.99
	4	12.7	33.8	$(33.8 / 4)$ $= 8.450$	8.4500	150.30
2538	1	6.5	33.8	$(33.8 / 4)$ $= 8.450$	8.4250	77.15
	2	4.6	33.6	$(33.6 / 4)$ $= 8.400$	8.5125	54.04
	3	9.8	34.5	$(34.5 / 4)$ $= 8.625$	8.6750	112.97
	4	13.6	34.9	$(34.9 / 4)$ $= 8.725$	8.7750	154.99
2539	1	6.9	35.3	$(35.3 / 4)$ $= 8.825$	8.9000	77.53
	2	5.0	35.9	$(35.9 / 4)$ $= 8.975$	9.0375	55.33

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Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$\frac{Y}{CMA} \times 100$
	3	10.4	36.4	$(36.4 / 4)$ = 9.100	9.1125	114.13
	4	14.1	36.5	$(36.5 / 4)$ = 9.125	9.1875	153.47
2540	1	7.0	37.0	$(37.0 / 4)$ = 9.250	9.3000	75.27
	2	5.5	37.4	$(37.4 / 4)$ = 9.350	9.4625	58.12
	3	10.8	38.3	$(38.3 / 4)$ = 9.575	9.5875	112.65
	4	15.0	38.4	$(38.4 / 4)$ = 9.600	9.6250	155.84
2541	1	7.1	38.6	$(38.6 / 4)$ = 9.650	9.6875	73.29
	2	5.7	38.9	$(38.9 / 4)$ = 9.725	9.6625	58.99
	3	11.1	38.4	$(38.4 / 4)$ = 9.600	9.7125	114.29
	4	14.5	39.3	$(39.3 / 4)$ = 9.825	9.8875	146.65
2542	1	8.0	39.8	$(39.8 / 4)$ = 9.950	9.9875	80.10

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Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$\frac{Y}{CMA} \times 100$
	2	6.2	40.1	(40.1 / 4) =10.025	10.0750	61.54
	3	11.4	40.5	(40.5 / 4) = 10.125	-	-
	4	14.9			-	-

Calculation of Multiplicative Seasonal Index Using the Ratio-to-Moving Average Method

The calculated values of $\frac{Y}{CMA} \times 100$ are rewritten in the table below, and the average values for Quarters 1, 2, 3, and 4 are then computed.

	Quarter				
Year	1	2	3	4	
2537	-	-	117.99	150.30	
2538	77.15	54.04	112.97	154.99	
2539	77.53	55.33	114.13	153.47	
2540	75.27	58.12	112.65	155.84	
2541	73.29	58.99	114.29	146.65	
2542	80.10	61.54	-	-	
Total	383.34	288.02	572.03	761.25	
Average	76.67	57.60	114.41	152.25	400.93
Seasonal Index (S%) (Adjusted)	$\frac{76.67}{400.93} \times 400$ = 76.49	$\frac{57.60}{400.93} \times 400$ = 57.47	$\frac{114.41}{400.93} \times 400$ = 114.14	$\frac{152.25}{400.93} \times 400$ = 151.90	400

- **The seasonal index for Quarter 1 equals 76.49**
which means that the effect of seasonal variation causes sales in Quarter 1 to be 23.51% **below normal** ($100 - 76.49 = 23.51$).

- **The seasonal index for Quarter 2 equals 57.47**
which means that the effect of seasonal variation causes sales in Quarter 2 to be 42.53% **below normal** ($100 - 57.47 = 42.53$).
- **The seasonal index for Quarter 3 equals 114.14**
which means that the effect of seasonal variation causes sales in Quarter 3 to be 14.14% **above normal** ($114.14 - 100 = 14.14$).
- **The seasonal index for Quarter 4 equals 151.90**
which means that the effect of seasonal variation causes sales in Quarter 4 to be 51.90% **above normal** ($151.90 - 100 = 51.90$).
- **Estimation of Additive Seasonal Variation Using the Ratio-to-Moving Average Method** The procedure can be summarized as follows:
 - 1) Isolate the trend component (T) from the original data **using the moving average method**

Suppose that

The number of periods in the moving average = the length of one complete seasonal cycle.

- If the annual data **consists of 4 periods** (4 quarters per year), use a **4-period moving average**.
- If the annual data **consists of 12 periods** (12 months per year), use a **12-period moving average**.

Note: If the number of **periods** in the moving average is even, a **centered moving average** must be calculated.

2) **Subtract the calculated moving averages from the corresponding original data (Y) to obtain ($Y - CMA$)**

3) Calculate the average of the results from Step 2

4) Verify that **the sum of the period averages** from Step 3 equals zero for one complete seasonal cycle

- If **the sum of averages** equals 0, the average for each period represents the "**additive seasonal index**"

- If **the sum of averages does not equal 0**, the averages for each period must be adjusted. Divide **the sum of the averages by the total number of periods in one seasonal cycle**

(divide by 4 if there are 4 quarters per year, or divide by 12 if there are 12 months per year), and subtract this value from the original average for each period. The adjusted **averages will sum to zero**, and these values represent the desired **additive seasonal index**.

Example The sales manager of a department store has **collected quarterly sales data** (every 3 months) from B.E. 2537–2542. Demonstrate how to **calculate the additive seasonal index** using **the ratio-to-moving average method**.

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Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$Y - CMA$
2537	1	6.7			-	-
	2	4.6			-	-
	3	10.0	34.0	$(34.0 / 4) = 8.500$	8.4750	1.5250
	4	12.7	33.8	$(33.8 / 4) = 8.450$	8.4500	4.2500
2538	1	6.5	33.8	$(33.8 / 4) = 8.450$	8.4250	1.9250
	2	4.6	33.6	$(33.6 / 4) = 8.400$	8.5125	3.9125
	3	9.8	34.5	$(34.5 / 4) = 8.625$	8.6750	1.1250
	4	13.6	34.9	$(34.9 / 4) = 8.725$	8.7750	4.8250
2539	1	6.9	35.3	$(35.3 / 4) = 8.825$	8.9000	2.0000
	2	5.0	35.9	$(35.9 / 4) = 8.975$	9.0375	4.0375
	3	10.4	36.4	$(36.4 / 4) = 9.100$	9.1125	1.2875

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Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$Y - CMA$
	4	14.1	36.5	$(36.5 / 4) =$ 9.125	9.1875	4.9125
2540	1	7.0	37.0	$(37.0 / 4) =$ 9.250	9.3000	2.3000
	2	5.5	37.4	$(37.4 / 4) =$ 9.350	9.4625	3.9625
	3	10.8	38.3	$(38.3 / 4) =$ 9.575	9.5875	1.2125
	4	15.0	38.4	$(38.4 / 4) =$ 9.600	9.6250	5.3750
2541	1	7.1	38.6	$(38.6 / 4) =$ 9.650	9.6875	2.5875
	2	5.7	38.9	$(38.9 / 4) =$ 9.725	9.6625	3.9625
	3	11.1	38.4	$(38.4 / 4) =$ 9.600	9.7125	1.3875
	4	14.5	39.3	$(39.3 / 4) =$ 9.825	9.8875	4.6125
2542	1	8.0	39.8	$(39.8 / 4) =$ 9.950	9.9875	1.9875
	2	6.2	40.1	$(40.1 / 4)$ =10.025	10.0750	3.8750

Year	Quarter	Sales (Y) (Million Baht)	4- Period Moving Total	4-Period Moving Average	Centered MA (CMA)	$Y - CMA$
	3	11.4	40.5	$(40.5 / 4) = 10.125$	-	-
	4	14.9			-	-

Calculation of Additive Seasonal Index Using the Ratio-to-Moving Average Method

Reorganize the calculated $(Y - CMA)$ values in the table below and find the average for quarters 1, 2, 3, and 4.

	Quarter				
Year	1	2	3	4	
2537	-	-	1.5250	4.2500	
2538	-1.9250	-3.9125	1.1250	4.8250	
2539	-2.0000	-4.0375	1.2875	4.9125	
2540	-2.3000	-3.9625	1.2125	5.3750	
2541	-2.5875	-3.9625	1.3875	4.6125	
2542	-1.9875	-3.8750	-	-	
Total	-10.8000	-19.7500	6.5375	23.9750	
Average	-2.1600	-3.9500	1.3075	4.7950	Total = -0.0075
Seasonal Index (S) (Adjusted)	$-2.1600 - \left(\frac{-0.0075}{4}\right)$ = -2.158	$-3.9500 - \left(\frac{-0.0075}{4}\right)$ = -3.948	$1.3075 - \left(\frac{-0.0075}{4}\right)$ = 1.309	$4.7950 - \left(\frac{-0.0075}{4}\right)$ = 4.797	Total = 0

- **The seasonal index for Quarter 1 equals -2.158**
which means that the effect of seasonal variation causes sales in Quarter 1 to be **2.158 million baht below normal**.
- **The seasonal index for Quarter 2 equals -3.948**
which means that the effect of seasonal variation causes sales in Quarter 2 to be **3.948 million baht below normal**.
- **The seasonal index for Quarter 3 equals 1.309**
which means that the effect of seasonal variation causes sales in Quarter 3 to be **1.309 million baht above normal**.

• **The seasonal index for Quarter 4 equals 4.797**

which means that the effect of seasonal variation causes sales in Quarter 4 to be **4.797 million baht above normal**.

Testing for Seasonal Influence Using the Kruskal-Wallis Test

Kruskal-Wallis Test (H-Test)

The Kruskal-Wallis test is a nonparametric statistical analysis method used to test whether the central tendencies of three or more populations $k \geq 3$ are different, when the k populations are independent of each other. The dependent variable is quantitative data measured at the interval or ratio level, while the independent variable is categorical data measured at the nominal or ordinal level with three or more groups.

Characteristics of Nonparametric Statistical Analysis:

- The data used for analysis can have lower quality measurement scales
- The data do not need to follow a normal distribution
- The analysis can be performed with small sample sizes
- The efficiency of the results obtained from the analysis is lower than that of parametric statistics

• **Kruskal-Wallis Testing Procedure**

1) State the hypotheses for testing

$$H_0 : M_1 = M_2 = \dots = M_k$$

(The medians of the k populations are not different) or
(the time series has no seasonal variation component)

$$H_1 : \textbf{At least one population median differs from the others}$$

(the time series has a seasonal variation component)

where $M_1, M_2, \dots, M_k$ represent the medians of populations 1, 2, ..., k ,

k = number of groups or number of seasons in one cycle;

$$k \geq 3$$

2) Calculate the test statistic

$$H = \left[\frac{12}{n(n+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right] - 3(n+1)$$

where: R_j = **sum** of ranks of observations in **group** j ;

$$j = 1, 2, \dots, k$$

n_j = sample size in group j

$$n = n_1 + n_2 + \dots + n_k$$

3) Decision

- **Reject** H_0 **and accept** H_1 : if $H > \chi^2_{\alpha, d.f.=k-1}$
- **Accept** H_0 : if $H \leq \chi^2_{\alpha, d.f.=k-1}$

Summary of Steps for Testing Seasonal Influence Using the Kruskal-Wallis Method

1. Estimate the time series trend for quarterly or monthly data using the Moving Average Method. **Use a 4-period moving average for quarterly data** and a **12-period moving average for monthly data**. When using an even number of periods, a centered moving average (CMA) must be calculated.

2. Divide the time series (Y) by the estimated trend value from Step 1 and multiply by 100. This represents $S \times I$ as follows:

$$S \times I = \frac{Y}{CMA} \times 100\%$$

Rank the variable data from the lowest value (rank 1) to the highest, continuing until all observations are ranked.

3. **Find the sum of ranks for group 1**, denoted as R_1

Find the sum of ranks for group 2, denoted as R_2

⋮

Find the sum of ranks for group k, denoted as R_k

4. Calculate the H statistic using the formula

$$H = \left[\frac{12}{n(n+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right] - 3(n+1)$$

Example Using the quarterly sales data and the estimated trend values by the Centered Moving Average (CMA) method from pages 3-4, test whether the detrended quarterly sales data exhibit seasonal influence at a significance level of 0.05

Table 1: Detrended Quarterly Sales Data $\left(S = \frac{Y}{CMA} \times 100\% \right)$

* Numbers in parentheses represent the assigned ranks

	Quarter			
Year	1	2	3	4
2537	-	-	117.99 (15)	150.30 (17)
2538	77.15 (8)	54.04 (1)	112.97 (12)	154.99 (19)
2539	77.53 (9)	55.33 (2)	114.13 (13)	153.47 (18)
2540	75.27 (7)	58.12 (3)	112.65 (11)	155.84 (20)
2541	73.29 (6)	58.99 (4)	114.29 (14)	146.65 (16)
2542	80.10 (10)	61.54 (5)	-	-
<u>Sum of Ranks</u>	$R_1 = 40$	$R_2 = 15$	$R_3 = 65$	$R_4 = 90$
<u>Average Rank</u>	$\bar{R}_1 = \frac{R_1}{n_1} = \frac{40}{5} = 8$	$\bar{R}_2 = \frac{R_2}{n_2} = \frac{15}{5} = 3$	$\bar{R}_3 = \frac{R_3}{n_3} = \frac{65}{5} = 13$	$\bar{R}_4 = \frac{R_4}{n_4} = \frac{90}{5} = 18$

Where: R_1 = Sum of positions for Group 1 $\bar{R}_1$ = Average position of Group 1

DATE Q 3 4 Y 2537.

The following new variables are being created:

Name	Label
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YEAR_	YEAR, not periodic
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QUARTER_	QUARTER, period 4
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DATE_	Date. Format: "QQ YYYY"
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/COMPRESSED.
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DATE Q 3 4 Y 2537.

The following new variables are being created:

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Name Label

YEAR_ YEAR, not periodic

QUARTER_ QUARTER, period 4

DATE_ Date. Format: "QQ YYYY"

NPAR TESTS

/K-W=S BY QUARTER_(1 4)

/MEDIAN=S BY QUARTER_(1 4)

/MISSING ANALYSIS.

NPAR Tests

Kruskal-Wallis Test

Ranks		
	QUARTER, period 4	N
S	1	$n_1 = 5$
	2	$n_2 = 5$
	3	$n_3 = 5$
	4	$n_4 = 5$
	Total	$n = 20$
		Mean Rank
		$\bar{R}_1 = 8.00$
		$\bar{R}_2 = 3.00$
		$\bar{R}_3 = 13.00$
		$\bar{R}_4 = 18.00$

Test Statistics ^{a,b}	
	S
Chi-Square	$H = 17.857$
df	$k - 1 = 3$
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable: QUARTER, period 4

1) State the hypotheses

$$H_0: M_1 = M_2 = M_3 = M_4$$

(quarterly sales data **do not** exhibit seasonal influence) H_1 : At least one differs from the others;

(quarterly sales data exhibit seasonal influence)

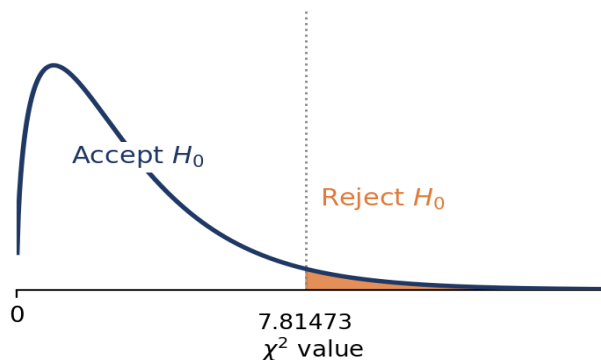
2) Set the significance level 0.05

3) Test statistic is

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$$\begin{aligned}
 H &= \left[\frac{12}{n(n+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right] - 3(n+1) \\
 &= \left[\frac{12}{n(n+1)} \left(\frac{R_1^2}{n_1} + \frac{R_2^2}{n_2} + \frac{R_3^2}{n_3} + \frac{R_4^2}{n_4} \right) \right] - 3(n+1) \\
 &= \left[\frac{12}{20(20+1)} \left(\frac{40^2}{5} + \frac{15^2}{5} + \frac{65^2}{5} + \frac{90^2}{5} \right) \right] - 3(20+1) \\
 &= \left[\frac{12}{20(21)} \times \frac{14150}{5} \right] - 63 \\
 &= 17.8571
 \end{aligned}$$



4) Critical region (or critical value from statistical table) is rejected H_0 if $H > \chi_{.05, k-1=3}^2$ $H > 7.81473$

4.1 Comparison of calculated H value with chi-square table value shows that

$H = 17.8571 > 7.81473$ Therefore, H_0 is rejected.

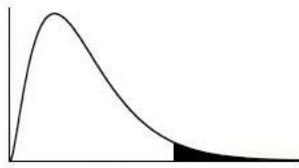
4.2 Comparison of *sig.* (P-Value) from output with significance level $\alpha = 0.05$ shows that $\text{Sig.}(P\text{-Value}) = 0.0005 < \alpha = 0.05 \therefore H_0$ is rejected

5) Decision (accept H_0 or reject H_0) ans. reject H_0 and accept H_1

6) Conclusion, the quarterly sales data exhibit significant seasonal influence at the 0.05 significance level.

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Percentage points of the χ^2 distributions

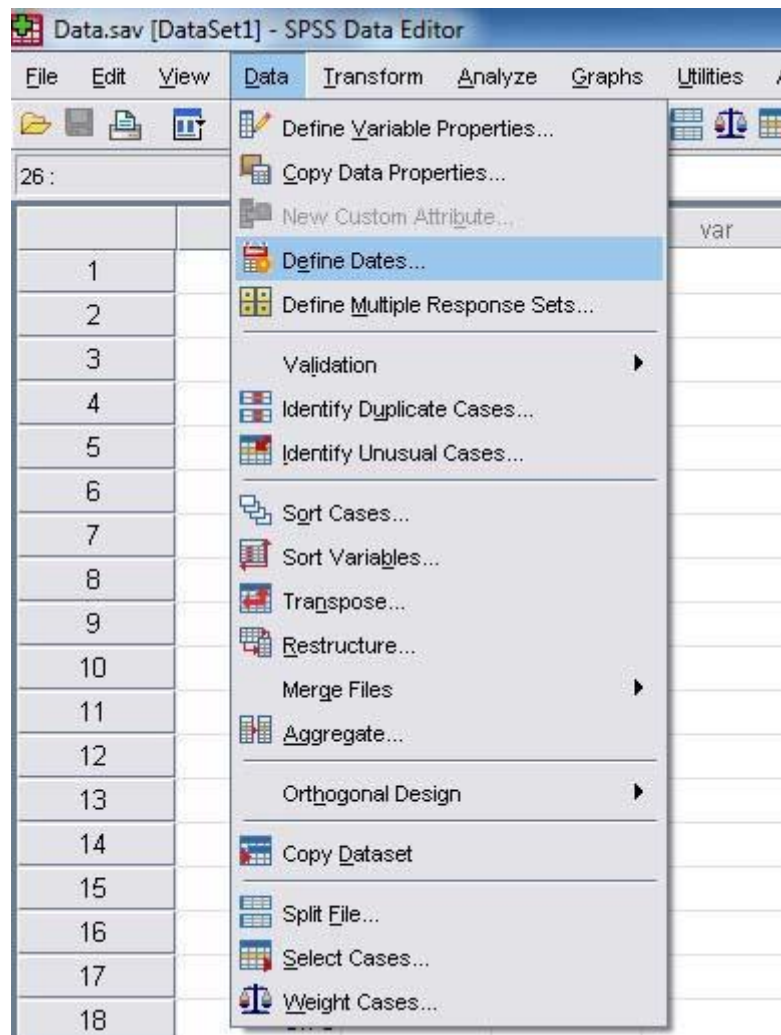
d.f	$\chi^2_{0.995}$	$\chi^2_{0.990}$	$\chi^2_{0.975}$	$\chi^2_{0.950}$	$\chi^2_{0.900}$	$\chi^2_{0.100}$	$\chi^2_{0.050}$	$\chi^2_{0.025}$	$\chi^2_{0.010}$	$\chi^2_{0.005}$
1	0.0000393	0.0001571	0.0009821	0.0039321	0.0157908	2.70554	3.84146	5.02389	6.63490	7.87944
2	0.0100251	0.0201007	0.0506356	0.102587	0.210721	4.60517	5.99146	7.37776	9.21034	10.5966
3	0.0717218	0.114832	0.215795	0.351846	0.584374	6.25139	7.81473	9.34840	11.3449	12.8382
4	0.206989	0.297109	0.484419	0.710723	1.063623	7.77944	9.48773	11.1433	13.2767	14.8603
5	0.411742	0.554298	0.831212	1.145476	1.61031	9.23636	11.0705	12.8325	15.0863	16.7496
6	0.675727	0.872090	1.237344	1.63538	2.20413	10.6446	12.5916	14.4494	16.8119	18.5476
7	0.989256	1.239042	1.68987	2.16735	2.83311	12.0170	14.0671	16.0128	18.4753	20.2777
8	1.344413	1.646497	2.17973	2.73264	3.48954	13.3616	15.5073	17.5345	20.0902	21.9550
9	1.734933	2.087901	2.70039	3.32511	4.16816	14.6837	16.9190	19.0228	21.6660	23.5894
10	2.15586	2.55821	3.24697	3.94030	4.86518	15.9872	18.3070	20.4832	23.2093	25.1882
11	2.60322	3.05348	3.81575	4.57481	5.57778	17.2750	19.6751	21.9200	24.7250	26.7568
12	3.07382	3.57057	4.40379	5.22603	6.30380	18.5493	21.0261	23.3367	26.2170	28.2995
13	3.56503	4.10692	5.00875	5.89186	7.04150	19.8119	22.3620	24.7356	27.6882	29.8195
14	4.07467	4.66043	5.62873	6.57063	7.78953	21.0641	23.6848	26.1189	29.1412	31.3193
15	4.60092	5.22935	6.26214	7.26094	8.54676	22.3071	24.9958	27.4884	30.5779	32.8013
16	5.14221	5.81221	6.90766	7.96165	9.31224	23.5418	26.2962	28.8454	31.9999	34.2672
17	5.69722	6.40776	7.56419	8.67176	10.0852	24.7690	27.5871	30.1910	33.4087	35.7185
18	6.26480	7.01491	8.23075	9.39046	10.8649	25.9894	28.8693	31.5264	34.8053	37.1565
19	6.84397	7.63273	8.90652	10.1170	11.6509	27.2036	30.1435	32.8523	36.1909	38.5823
20	7.43384	8.26040	9.59078	10.8508	12.4426	28.4120	31.4104	34.1696	37.5662	39.9968
21	8.03365	8.89720	10.28290	11.5913	13.2396	29.6151	32.6706	35.4789	38.9322	41.4011
22	8.64272	9.54249	10.9823	12.3380	14.0415	30.8133	33.9244	36.7807	40.2894	42.7957
23	9.26042	10.19572	11.6886	13.0905	14.8480	32.0069	35.1725	38.0756	41.6384	44.1813
24	9.88623	10.8564	12.4012	13.8484	15.6587	33.1962	36.4150	39.3641	42.9798	45.5585
25	10.5197	11.5240	13.1197	14.6114	16.4734	34.3816	37.6525	40.6465	44.3141	46.9279
26	11.1602	12.1981	13.8439	15.3792	17.2919	35.5632	38.8851	41.9232	45.6417	48.2899
27	11.8076	12.8785	14.5734	16.1514	18.1139	36.7412	40.1133	43.1945	46.9629	49.6449
28	12.4613	13.5647	15.3079	16.9279	18.9392	37.9159	41.3371	44.4608	48.2782	50.9934
29	13.1211	14.2565	16.0471	17.7084	19.7677	39.0875	42.5570	45.7223	49.5879	52.3356
30	13.7867	14.9535	16.7908	18.4927	20.5992	40.2560	43.7730	46.9792	50.8922	53.6720
40	20.7065	22.1643	24.4330	26.5093	29.0505	51.8051	55.7585	59.3417	63.6907	66.7660
50	27.9907	29.7067	32.3574	34.7643	37.6886	63.1671	67.5048	71.4202	76.1539	79.4900
60	35.5345	37.4849	40.4817	43.1880	46.4589	74.3970	79.0819	83.2977	88.3794	91.9517
70	43.2752	45.4417	48.7576	51.7393	55.3289	85.5270	90.5312	95.0232	100.425	104.215
80	51.1719	53.5401	57.1532	60.3915	64.2778	96.5782	101.879	106.629	112.329	116.321
90	59.1963	61.7541	65.6466	69.1260	73.2911	107.565	113.145	118.136	124.116	128.299
100	67.3276	70.0649	74.2219	77.9295	82.3581	118.498	124.342	129.561	135.807	140.169

Testing Seasonal Effects Using the Kruskal–Wallis Method with SPSS

Example: Testing for seasonal effects in **quarterly sales data**

0. Enter the data in the Data View window and define the date variable as follows:

Click Data → Define Dates...

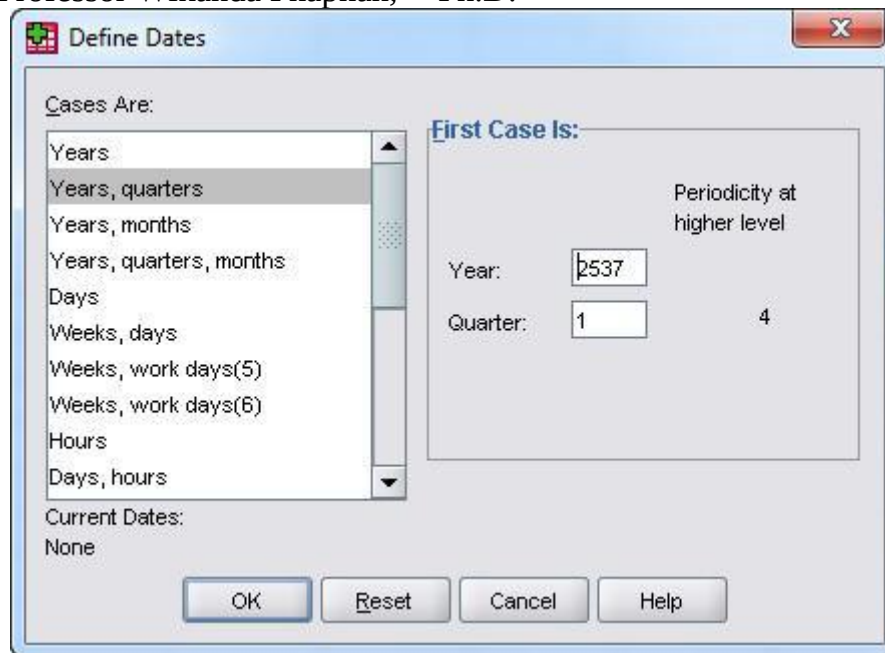


(1) Select **Year, Quarter** for quarterly data.

(2) Select **Year, Month** for monthly data.

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*Data.sav [DataSet1] - SPSS Data Editor

	Y	YEAR_	QUARTER_	DATE_	var
1	6.70	2537	1	Q1 2537	
2	4.60	2537	2	Q2 2537	
3	10.00	2537	3	Q3 2537	
4	12.70	2537	4	Q4 2537	
5	6.50	2538	1	Q1 2538	
6	4.60	2538	2	Q2 2538	
7	9.80	2538	3	Q3 2538	
8	13.60	2538	4	Q4 2538	
9	6.90	2539	1	Q1 2539	
10	5.00	2539	2	Q2 2539	
11	10.40	2539	3	Q3 2539	
12	14.10	2539	4	Q4 2539	

1. Estimating Trend Using Centered Moving Average ($Q_{150}; X=27$)

Method

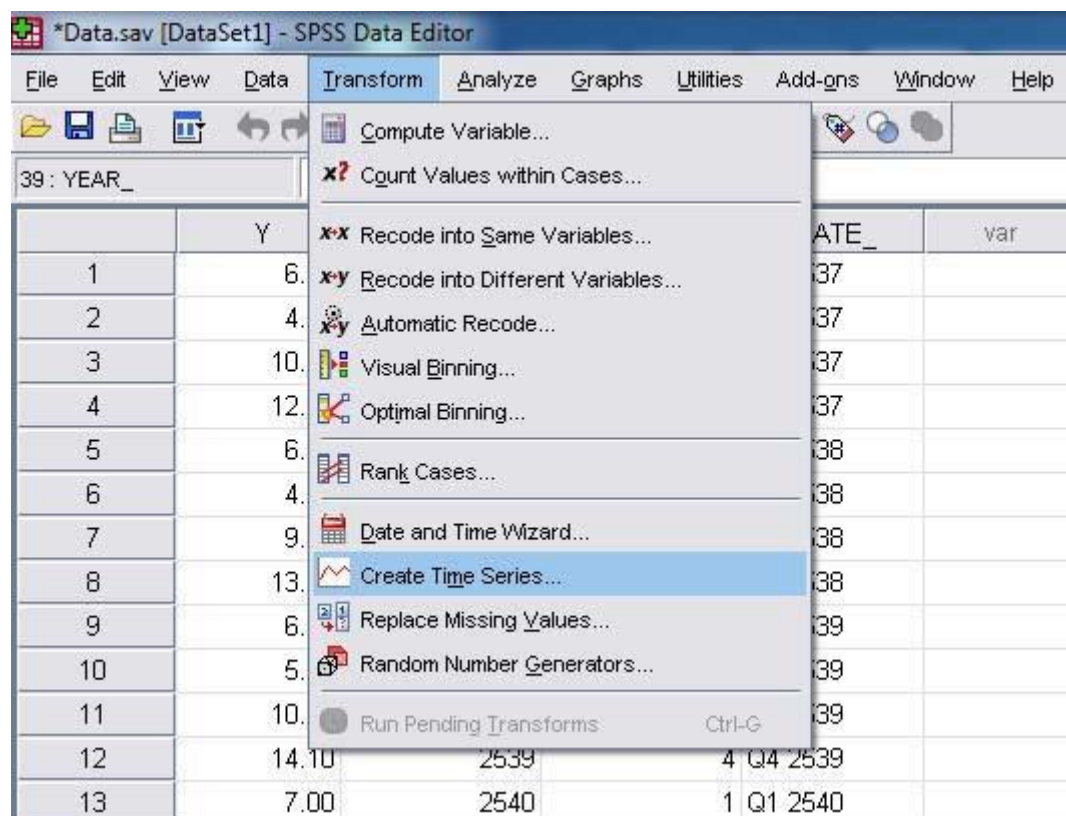
Use a 4-quarter moving average span because the data are quarterly (1 year = 4 quarters)

Commands: Transform → Create Time Series...

- Under Function, select Centered moving average
- In the Span box, enter 4 (since the data are quarterly).
- Under New Variable (s), move the variable y into the box and click OK
- A new variable named CMA will appear in the Data View window. This variable contains the values obtained from the **centered moving average** calculation.

Alternative Method: Calculating Centered Moving Average (CMA)

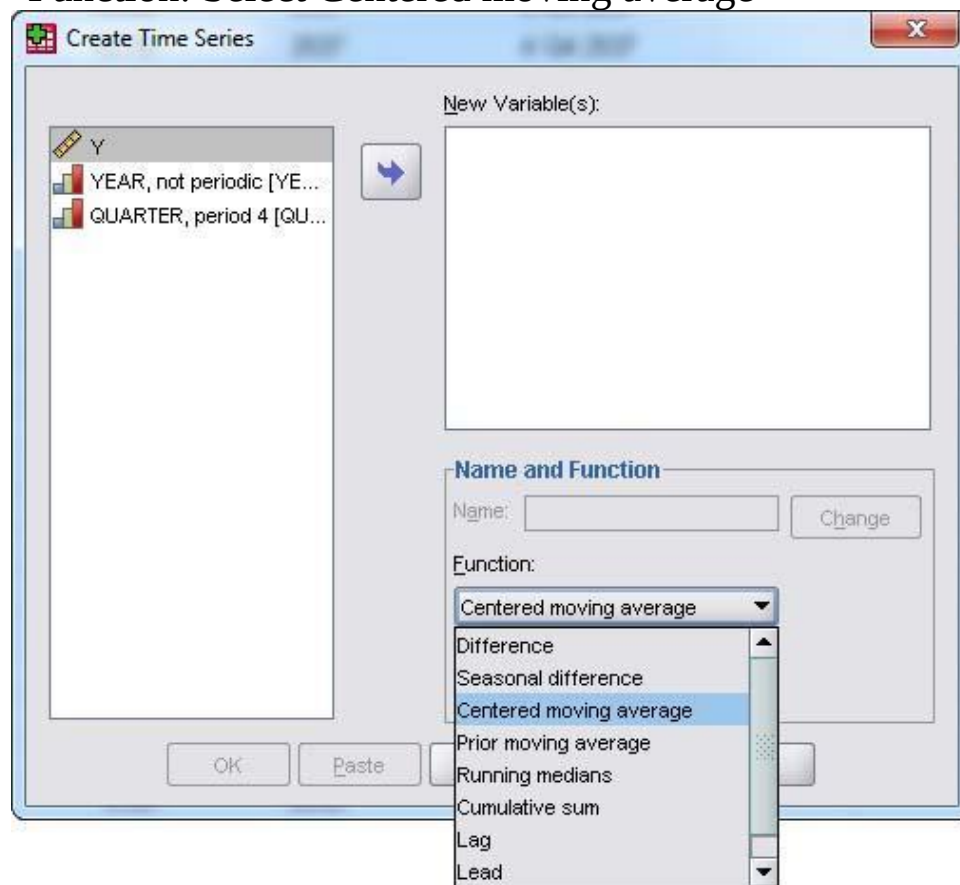
- Click Transform → Create Time Series...



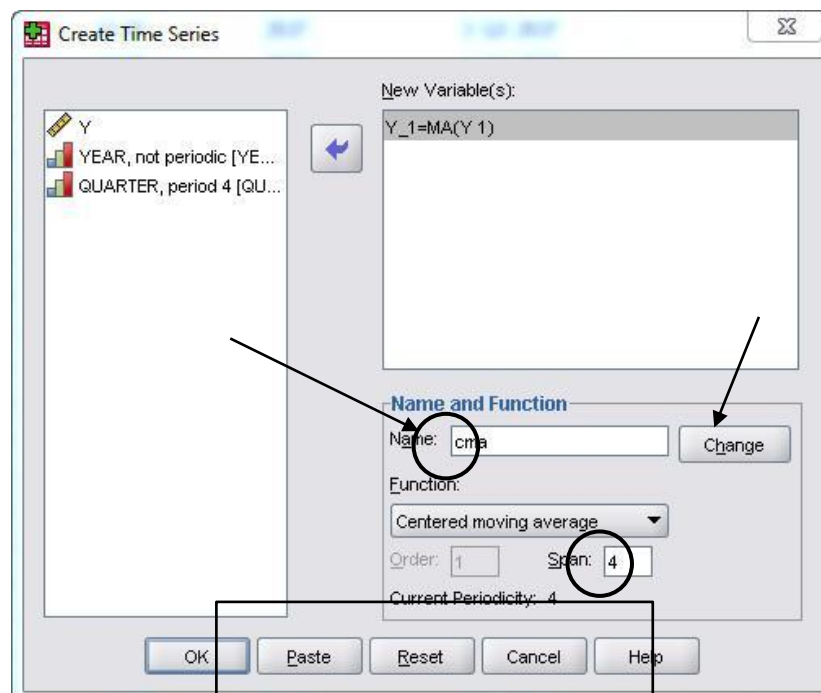
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- Function: Select Centered moving average



- In Span: enter 4 (for quarterly data).
- Under New Variable (s), move the variable into the box.

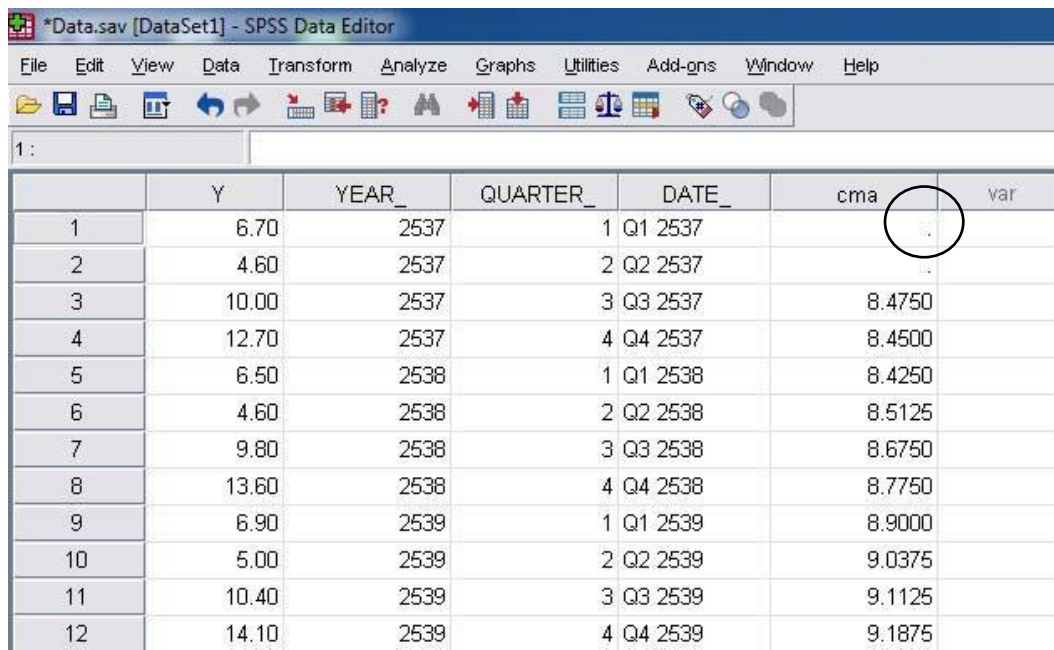


Rename y_1
to **cma** and
click **Change**

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- In Name and Function, type the name of the variable that will store the moving average values. In the Name: cma and click **Change**
- A new variable "cma" will appear in the Data window, which contains the calculated centered moving average values.



	Y	YEAR_	QUARTER_	DATE_	cma	var
1	6.70	2537	1	Q1 2537		
2	4.60	2537	2	Q2 2537		
3	10.00	2537	3	Q3 2537	8.4750	
4	12.70	2537	4	Q4 2537	8.4500	
5	6.50	2538	1	Q1 2538	8.4250	
6	4.60	2538	2	Q2 2538	8.5125	
7	9.80	2538	3	Q3 2538	8.6750	
8	13.60	2538	4	Q4 2538	8.7750	
9	6.90	2539	1	Q1 2539	8.9000	
10	5.00	2539	2	Q2 2539	9.0375	
11	10.40	2539	3	Q3 2539	9.1125	
12	14.10	2539	4	Q4 2539	9.1875	

2. Creating a New Variable Representing Detrended Data or Calculating the Ratio-to-Moving Average, this variable will be denoted as S

Commands: Transform → Compute

- Target Variable: Type s (the name of the variable to be created)
- Numeric Expansion: Enter the formula for calculating the seasonal index: $(y/cma) \times 100$ or $\frac{Y}{CMA} \times 100$ and click OK
- new variable s will appear in the Data window, representing the detrended data.

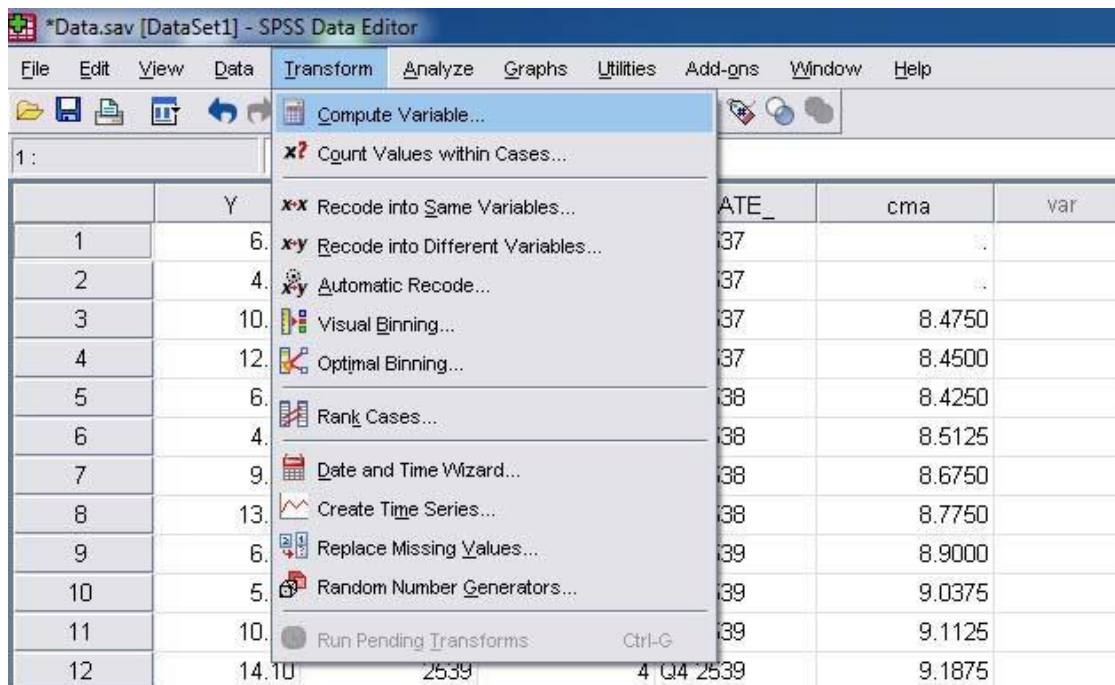
A new variable is created to represent the data with the trend component removed. This process involves calculating the ratio to

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the centered moving average. In this case, the resulting variable is denoted by S

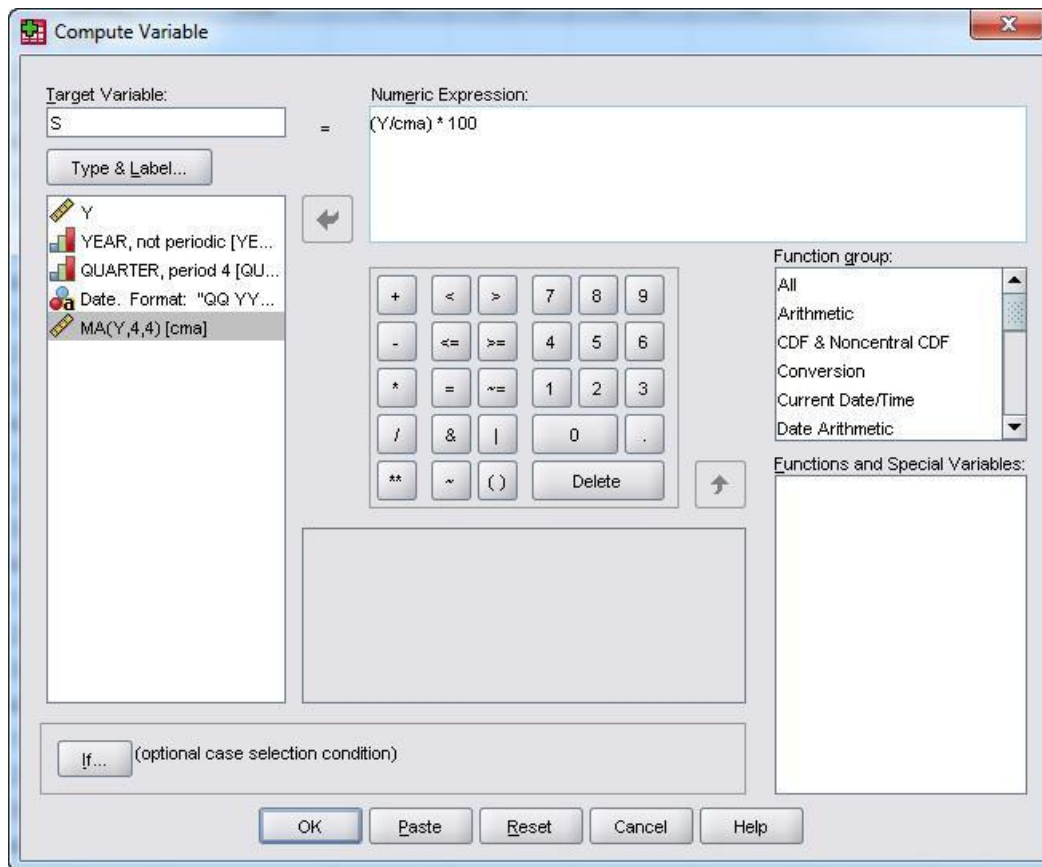
- Click Transform → Compute...



- Target Variable: Type S (the name of the variable to be created) and
- Numeric Expansion: Enter the formula for calculating the seasonal index: $(y/cma)*100$ or $\frac{Y}{CMA} \times 100$ and click **OK**

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- The Data window will display the new variable "S" representing the detrended data.

*Data.sav [DataSet1] - SPSS Data Editor

	Y	YEAR_	QUARTER_	DATE_	cma	S	var
1	6.70	2537	1	Q1 2537			
2	4.60	2537	2	Q2 2537			
3	10.00	2537	3	Q3 2537	8.4750	117.99	
4	12.70	2537	4	Q4 2537	8.4500	150.30	
5	6.50	2538	1	Q1 2538	8.4250	77.15	
6	4.60	2538	2	Q2 2538	8.5125	54.04	
7	9.80	2538	3	Q3 2538	8.6750	112.97	
8	13.60	2538	4	Q4 2538	8.7750	154.99	
9	6.90	2539	1	Q1 2539	8.9000	77.53	
10	5.00	2539	2	Q2 2539	9.0375	55.33	
11	10.40	2539	3	Q3 2539	9.1125	114.13	
12	14.10	2539	4	Q4 2539	9.1875	153.47	
13	7.00	2540	1	Q1 2540	9.3000	75.27	

3. Calculating the Kruskal-Wallis Test Statistic (H-test)

Commands: Analyze → Nonparametric Tests → K Independent Samples

- Target Variable List: **Select s** (the detrended data expressed as a percentage $\frac{Y}{CMA} \times 100\%$)

- Group Variable: **Select** quarter (1, 4)

Click Define Range... to specify the range of groups for testing

Set Minimum = 1 (representing Quarter 1, Group 1)

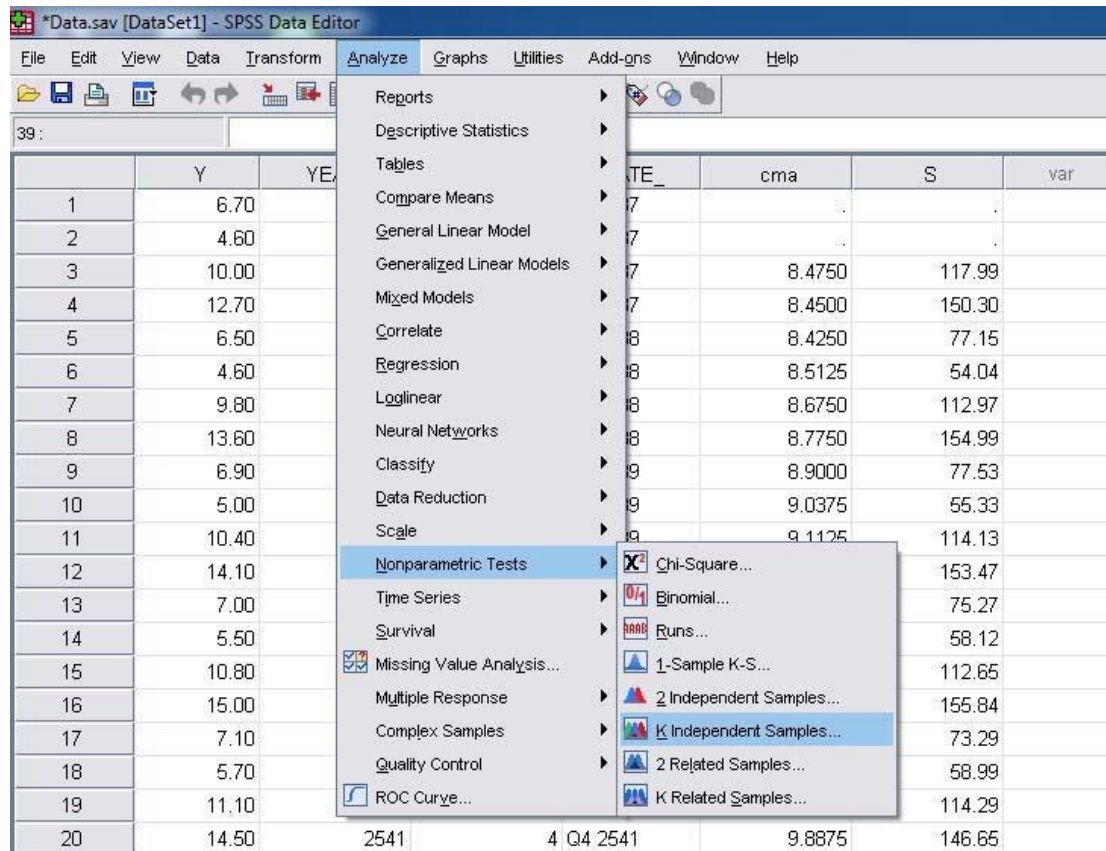
Set Maximum = 4 (representing Quarter 4, Group 4)

This tests whether the average detrended values from **Quarter 1** (Group 1) through **Quarter 4** (Group 4) differ significantly and click OK

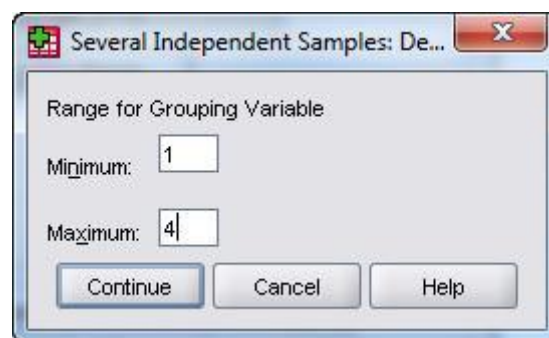
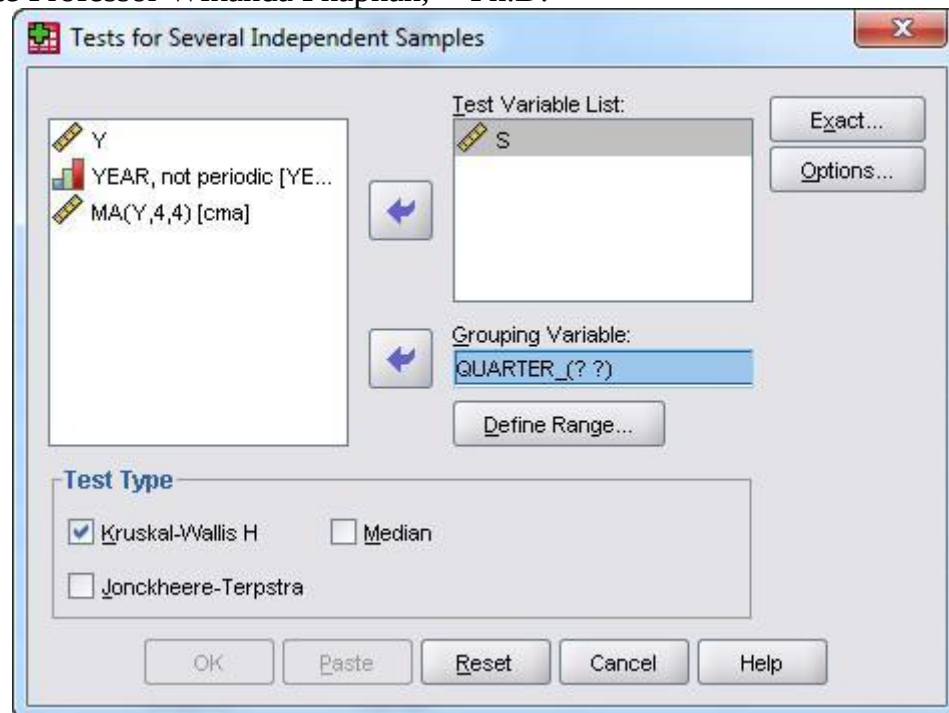
- The Output window will display the Kruskal-Wallis test statistic (H-test) and the p-value.

Calculation of the Kruskal–Wallis H-test statistic

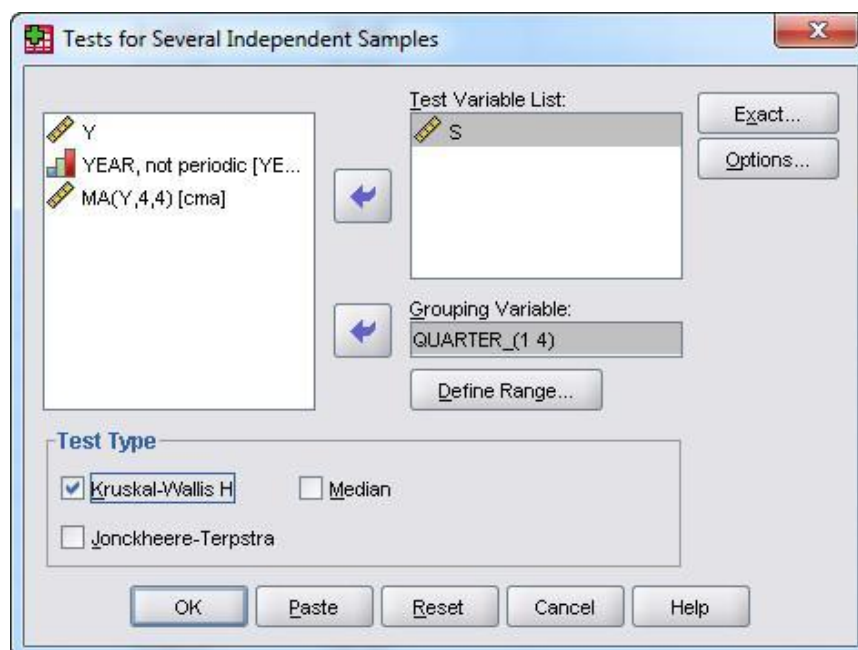
- Click Analyze → Nonparametric Tests → K Independent Samples...



- Test Variable List : Select variable S
(S the detrended data expressed as a percentage $\frac{Y}{CMA} \times 100\%$)
- Group Variable: Select quarter
- Click Define Range... and specify the boundaries of the Grouping Variable **Since the data are quarterly**, set Minimum = 1. and Maximum = 4. and Click **OK**



- Test Type ☒ Kruskal-Wallis H Click OK



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- The Output window will display the Kruskal-Wallis test statistic (H-test) and the p-value.

The screenshot shows the SPSS Output window for 'NPar Tests'. The left pane shows a tree view with 'Kruskal-Wallis Test' expanded, showing 'Title', 'Ranks', and 'Test Statistics'. The main pane displays the following tables:

NPar Tests

Kruskal-Wallis

Ranks

	QUARTER, period 4	N	Mean Rank
S	1	5	8.00
	2	5	3.00
	3	5	13.00
	4	5	18.00
	Total	20	

Test Statistics^{a,b}

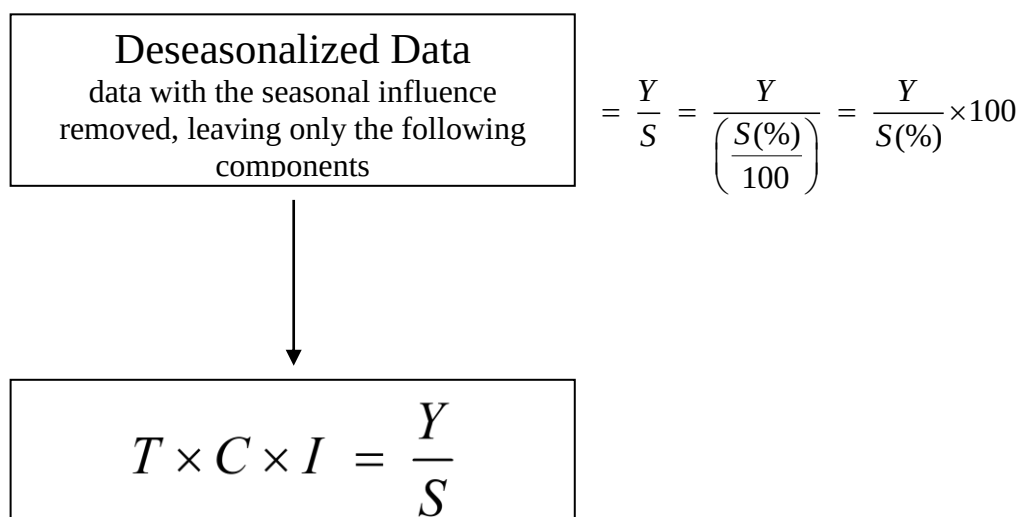
	S
Chi-Square	17.857
df	3
Asymp. Sig.	.000

a. Kruskal Wallis Test
b. Grouping Variable: QUARTER, period 4

- **Deseasonalized Data**

Deseasonalized data refers to data from which seasonal variation has been removed. This is accomplished by dividing the original data (Y) for each month or quarter by the corresponding seasonal index for that month or quarter, then multiplying by 100.

(Note: Deseasonalized data has the same units as Y)



(From the model: $Y = T \times S \times C \times I$)

● **Benefits of Deseasonalized Data**

1. Reveals true data movement: Shows the actual trends and patterns in the data after adjusting for seasonal influences
2. In planning and decision-making: Supports production planning and inventory management decisions

Example Using the quarterly sales data from a department store and **the seasonal indices calculated by the ratio-to-moving average method**, answer the following questions:

- 1) Calculate the deseasonalized data for quarters 1, 2, 3, and 4 of the year B.E. 2537

$$\begin{aligned}
 Q_{1/2537} : \quad \frac{Y}{S_{\text{ไตรมาส 1}}(\%)} \times 100 &= \frac{6.7}{76.49} \times 100 = 8.76 \quad \text{million baht} \\
 Q_{2/2537} : \quad \frac{Y}{S_{\text{ไตรมาส 2}}(\%)} \times 100 &= \frac{4.6}{57.47} \times 100 = 8.00 \quad \text{million baht} \\
 Q_{3/2537} : \quad \frac{Y}{S_{\text{ไตรมาส 3}}(\%)} \times 100 &= \frac{10.0}{114.14} \times 100 = 8.76 \quad \text{million baht} \\
 Q_{4/2537} : \quad \frac{Y}{S_{\text{ไตรมาส 4}}(\%)} \times 100 &= \frac{12.7}{151.90} \times 100 = 8.36 \quad \text{million baht}
 \end{aligned}$$

- 2) Demonstrate how to calculate the actual sales for the last quarter of B.E. 2542 for short-term planning purposes, excluding the influence of seasonal variation (i.e., after seasonal adjustment)

Calculate the deseasonalized data $\frac{Y}{S(\%)} \times 100$ for Quarter 4 of B.E.

2542

$$Q_{4/2542} : \quad \frac{Y}{S_{\text{ไตรมาส 4}}(\%)} \times 100 = \frac{14.9}{151.90} \times 100 = 9.81 \quad \text{million baht}$$

Therefore, in Quarter 4 of B.E. 2542, the actual sales after removing seasonal influence equal 9.81 million baht.

Using the quarterly sales data, the seasonally adjusted data can be computed as follows.

Year	Quarter	Seasonal Index (S%) Ratio-to-Moving Average Method	Y	$T \times C \times I = \frac{Y}{S(\%)} \times 100$ (Deseasonalized Data)
2537	1	76.49	6.7	8.76
	2	57.47	4.6	8.00
	3	114.14	10.0	8.76
	4	151.9	12.7	8.36
2538	1	76.49	6.5	8.50
	2	57.47	4.6	8.00
	3	114.14	9.8	8.59
	4	151.9	13.6	8.95
2539	1	76.49	6.9	9.02
	2	57.47	5.0	8.70
	3	114.14	10.4	9.11
	4	151.9	14.1	9.28
2540	1	76.49	7.0	9.15
	2	57.47	5.5	9.57
	3	114.14	10.8	9.46

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Year	Quarter	Seasonal Index(S%) Ratio-to- Moving Average Method	Y	$T \times C \times I = \frac{Y}{S(\%)} \times 100$ (Deseasonalized Data)
	4	151.9	15.0	9.87
2541	1	76.49	7.1	9.28
	2	57.47	5.7	9.92
	3	114.14	11.1	9.72
	4	151.9	14.5	9.55
2542	1	76.49	8.0	10.46
	2	57.47	6.2	10.79
	3	114.14	11.4	9.99
	4	151.9	14.9	9.81

Use seasonally adjusted data $T \times C \times I$ to estimate the linear trend equation.

```
* Curve Estimation.
TSET MXNEWVAR=1.
PREDICT THRU END.
CURVEFIT
/VARIABLES=TCI
/CONSTANT
/MODEL=LINEAR
/PRINT ANOVA
/PLOT FIT
/SAVE=PRED
/ID=DATE_.
```

Curve Fit

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Model Description

Model Name	MOD_1
Dependent Variable	1
Equation	1
Independent Variable	Case sequence
Constant	Included
Variable Whose Values Label Observations in Plots	Date. Format: "QQ YYYY"

Case Processing Summary

	N
Total Cases	24
Excluded Cases ^a	0
Forecasted Cases	0
Newly Created Cases	0

a. Cases with a missing value in any variable are excluded from the analysis.

Variable Processing Summary

		Variables
		Dependent
		TCI
Number of Positive Values		24
Number of Zeros		0
Number of Negative Values		0
Number of Missing Values	User-Missing	0
	System-Missing	0

TCI**Linear****Model Summary**

R	R Square	Adjusted R Square	Std. Error of the Estimate
.886	.785	.775	.340

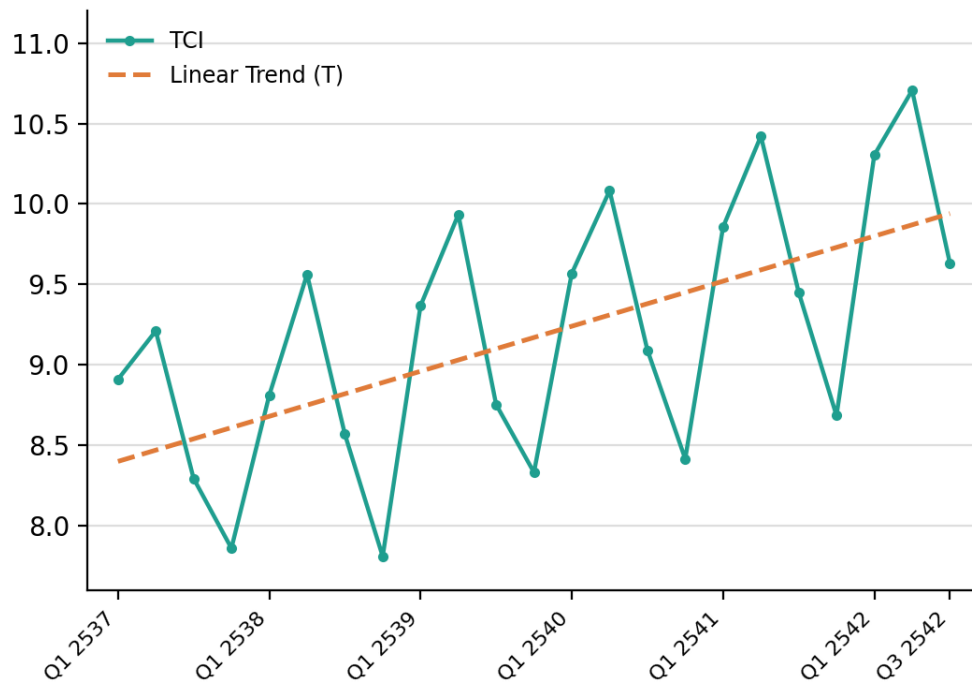
ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	9.308	1	9.308	80.450	.000
Residual	2.545	22	.116		
Total	11.853	23			

Coefficients

	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
Case Sequence	.090	.010	.886	8.969	.000
(Constant)	8.109	.143		56.578	.000

5.2



Estimation of Cyclical (C) and Irregular (I) Variations

The Residual Method is employed to systematically eliminate time series components—namely, trend (T), seasonal variation (S), and irregular variation (I) from the original data, leaving the desired cyclical component (C) of the time series.

- **The estimation of cyclical variation** can be summarized in the following steps:

- 1) Remove the trend component (T) and seasonal index (S) from the time series data (Y) by dividing the original data Y by both T and S

$$(Y = T \times S \times C \times I)$$

$$\frac{Y}{T \times \left(\frac{S(\%)}{100} \right)} \times 100 = \frac{T \times S \times C \times I}{T \times \left(\frac{S(\%)}{100} \right)} \times 100 = (C \times I)(\%)$$

- 2) Eliminate the irregular component (I) from the results obtained in 1) using **the weighted moving average method**. The weights are derived from binomial distribution coefficients, as follows:

- **For a 3-period moving average:** use weights of 1:2:1 **and divide by 4**
- **For a 4-period moving average:** use weights of 1:3:3:1 **and divide by 8** (Note: when using an even number of periods for the moving average, a centered moving average must be calculated)
- **For a 5-period moving average:** use weights of 1:4:6:4:1 **and divide by 16**

Upon completing steps 1) and 2), the resulting data will contain the desired cyclical variation component (C)

- **The estimation of cyclical variation** using weighted averaging with weights of 1 : 2 : 1

Time period	$(C \times I)(\%) = \frac{Y}{T \times S} \times 100$	$C(\%) = \text{Weighted moving average}$ $(C \times I)(\%)$
1	a	-
2	b	$\frac{1a + 2b + 1c}{4}$
3	c	$\frac{1b + 2c + 1d}{4}$
4	d	$\frac{1c + 2d + 1e}{4}$
5	e	-

• **Estimation of Cyclical (C) and Irregular (I) Variations**

multiplicative
model:
$$Y = T \times S \times C \times I$$
, by division.

1. **If Y, T, and S are known, and we want to study the influence of C × I**

$$(C \times I)(\%) = \frac{Y}{T \times S} \times 100 = \frac{Y}{\left(T \times \frac{S(\%)}{100}\right)} \times 100$$

2. **If Y, T, S, and I are known, and we want to study the influence of C**

$$C(\%) = \frac{Y}{T \times S \times I} \times 100 = \frac{Y}{\left(T \times \frac{S(\%)}{100} \times \frac{I(\%)}{100}\right)} \times 100$$

3. **If Y, T, S, and C are known, and we want to study the influence of I**

$$I(\%) = \frac{Y}{T \times S \times C} \times 100 = \frac{Y}{\left(T \times \frac{S(\%)}{100} \times \frac{C(\%)}{100}\right)} \times 100$$

4. **If deseasonalized data (T × C × I) and T are known, and we want to study C × I:**

$$(C \times I)(\%) = \frac{T \times C \times I}{T} \times 100 = \frac{\text{Seasonally adjusted data}}{\text{Trend}} \times 100$$

5. **If deseasonalized data (T × C × I), T, and I are known, and we want to study C**

$$C(\%) = \frac{T \times C \times I}{T \times I} \times 100 = \frac{T \times C \times I}{\left(T \times \frac{I(\%)}{100}\right)} \times 100$$

6. **If deseasonalized data (T × C × I), T, and C are known, and we want to study I**

$$I(\%) = \frac{T \times C \times I}{T \times C} \times 100 = \frac{T \times C \times I}{\left(T \times \frac{C(\%)}{100}\right)} \times 100$$

- **Estimation of the Cyclical (C) and Irregular (I) Components**

additive model:

$$Y = T + S + C + I$$

by subtraction as follows:

1. If Y, T, and S are known, and we want to study the influence of C + I

$$C + I = Y - T - S$$

2. If Y, T, S, and I are known, and we want to study the influence of C

$$C = Y - T - S - I$$

3. If Y, T, S, and C are known, and we want to study the influence of I

$$I = Y - T - S - C$$

4. If deseasonalized data (T + C + I) and T are known, and we want to study C + I

$$C + I = (T + C + I) - T = \text{Seasonally Adjusted Data} - T$$

5. If deseasonalized data (T + C + I), T, and I are known, and we want to study C

$$C = (T + C + I) - T - I = \text{Seasonally Adjusted Data} - T - I$$

6. If deseasonalized data (T + C + I), T, and C are known, and we want to study I

$$I = (T + C + I) - T - C = \text{Seasonally Adjusted Data} - T - C$$

Table: Estimation of Cyclical Variation (C) and Irregular Variation (I)

Example: Using quarterly sales data, calculate the influence of $C \times I$ for each quarter from B.E. 2537 - 2542

Year	Quarter	Y	X	$\hat{Y} = 8.11 + 0.09X$	S(%)	$(C \times I)(\%) = \frac{Y}{\left(T \times \frac{S(\%)}{100}\right)} \times 100$
2537	1	6.7	1	8.20	76.49	106.82
	2	4.6	2	8.29	57.47	96.55
	3	10.0	3	8.38	114.14	104.55
	4	12.7	4	8.47	151.9	98.71
2538	1	6.5	5	8.56	76.49	99.27
	2	4.6	6	8.65	57.47	92.53
	3	9.8	7	8.74	114.14	98.24
	4	13.6	8	8.83	151.9	101.40
2539	1	6.9	9	8.92	76.49	101.13
	2	5.0	10	9.01	57.47	96.56
	3	10.4	11	9.10	114.14	100.13
	4	14.1	12	9.19	151.9	101.01
2540	1	7.0	13	9.28	76.49	98.62
	2	5.5	14	9.37	57.47	102.14
	3	10.8	15	9.46	114.14	100.02
	4	15.0	16	9.55	151.9	103.40
2541	1	7.1	17	9.64	76.49	96.29
	2	5.7	18	9.73	57.47	101.93
	3	11.1	19	9.82	114.14	99.03
	4	14.5	20	9.91	151.9	96.32
2542	1	8.0	21	10.00	76.49	104.59
	2	6.2	22	10.09	57.47	106.92
	3	11.4	23	10.18	114.14	98.11
	4	14.9	24	10.27	151.9	95.51

Note: $S(\%)$ represents the seasonal index in percentage form **obtained using the ratio-to-moving-average method.**

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$Q_{1/2537} : C \times I = \underline{106.82} \Rightarrow$ In Quarter 1 of 2537, the sales data(Y) was influenced by C and I , causing sales to be 6.82% **higher** than normal.

$Q_{2/2537} : C \times I = \underline{96.55} \Rightarrow$ In Quarter 2 of 2537, the sales data(Y) was influenced by C and I , causing sales to be 3.45% **lower** than normal.

Table: Estimation of Cyclical Variation (C)

Example: Using quarterly sales data, calculate the influence of C for each quarter from B.E. 2537 – 2542

Year	Quarter	Y	X	$\hat{Y} = 8.11 + 0.09X$ (T)	$S(\%)$	$(C \times I)(\%)$	(C)(%) ($C \times I$)(%) Use a weighted moving average 1 : 2 : 1	$I(\%) = \frac{[C \times I]\%}{C(\%)} \times 100$
2537	1	6.7	1	8.20	76.49	106.82	-	-
	2	4.6	2	8.29	57.47	96.55	101.1175	95.48
	3	10.0	3	8.38	114.14	104.55	101.0900	103.42
	4	12.7	4	8.47	151.9	98.71	100.3100	98.40
2538	1	6.5	5	8.56	76.49	99.27	97.4450	101.87
	2	4.6	6	8.65	57.47	92.53	95.6425	96.75
	3	9.8	7	8.74	114.14	98.24	97.6025	100.65
	4	13.6	8	8.83	151.9	101.40	100.5425	100.85
2539	1	6.9	9	8.92	76.49	101.13	100.0550	101.07
	2	5.0	10	9.01	57.47	96.56	98.5950	97.94
	3	10.4	11	9.10	114.14	100.13	99.4575	100.68
	4	14.1	12	9.19	151.9	101.01	100.1925	100.82
2540	1	7.0	13	9.28	76.49	98.62	100.0975	98.52
	2	5.5	14	9.37	57.47	102.14	100.7300	101.40
	3	10.8	15	9.46	114.14	100.02	101.3950	98.64
	4	15.0	16	9.55	151.9	103.40	100.7775	102.60
2541	1	7.1	17	9.64	76.49	96.29	99.4775	96.80

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Year	Quarter	Y	X	$\hat{Y} = 8.11 + 0.09X$ (T)	S(%)	(C × I)(%)	(C)(%) (C × I)(%) Use a weighted moving average 1 : 2 : 1	$I(\%) = \frac{[C \times I]\%}{C(\%)} \times 100$
2542	2	5.7	18	9.73	57.47	101.93	99.7950	102.14
	3	11.1	19	9.82	114.14	99.03	99.0775	99.95
	4	14.5	20	9.91	151.9	96.32	99.0650	97.23
	1	8.0	21	10.00	76.49	104.59	103.1050	101.44
	2	6.2	22	10.09	57.47	106.92	104.1350	102.67
	3	11.4	23	10.18	114.14	98.11	99.6625	98.44
	4	14.9	24	10.27	151.9	95.51	-	-

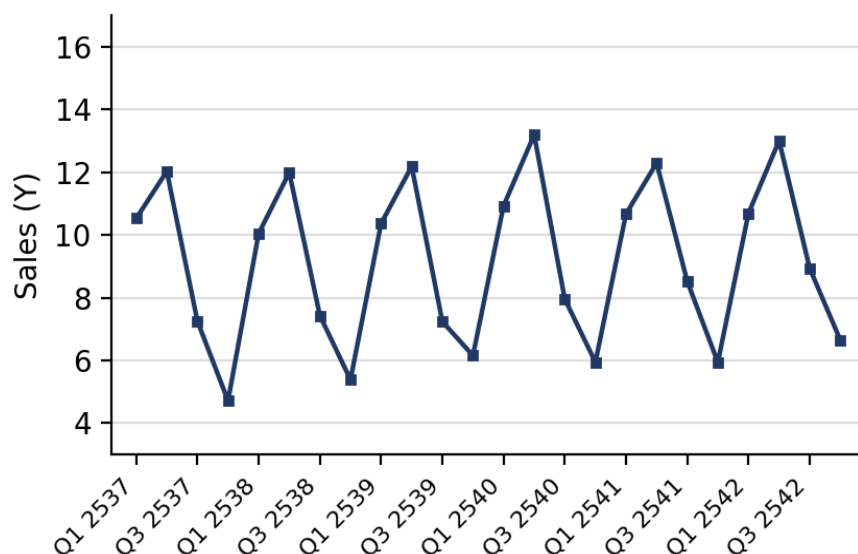


Figure 1 Quarterly sales data from Quarter 1, B.E. 2537 – to Quarter 4, B.E. 2542

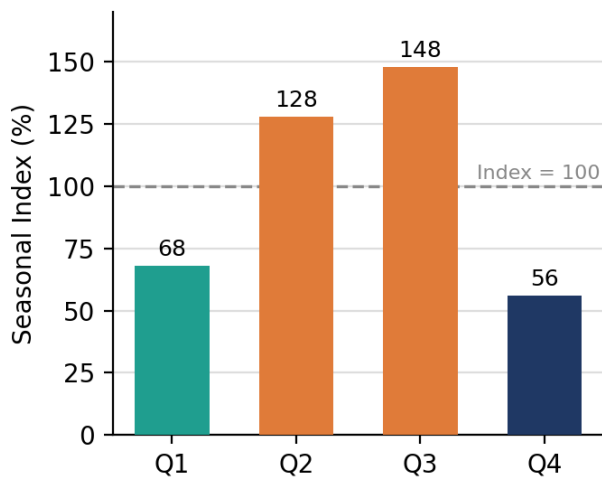


Figure 2 Seasonal variation in quarterly sales.

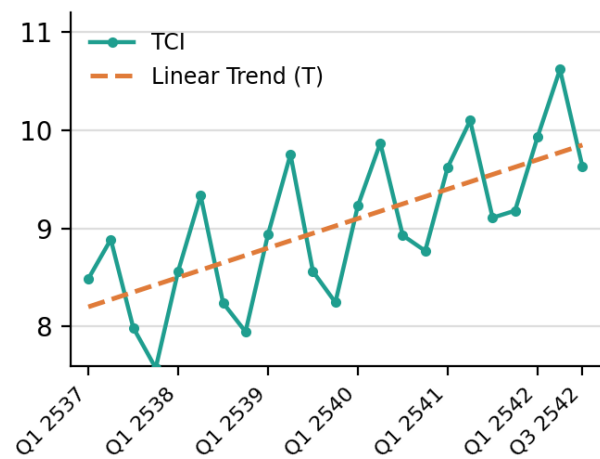


Figure 3 The TCI components and the linear trend (T).

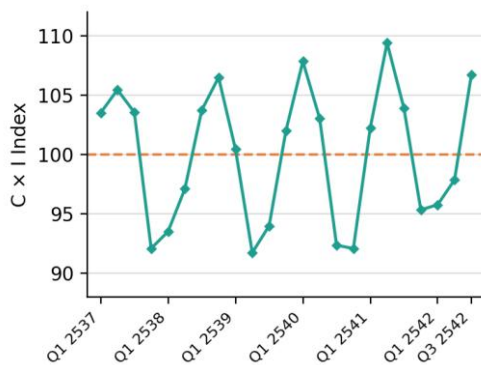


Figure 4 Variation in the components $C \times I$

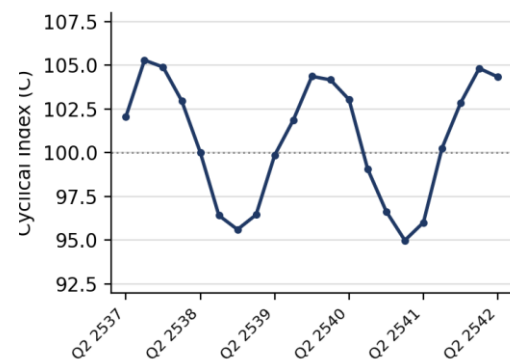


Figure 5 Cyclical variation (C).

5.3 Forecasting

Forecasting can be performed using the **multiplicative model**: $Y = T \times S \times C \times I$, which can be approached in two ways:

1) If all components are known (T , S , C , and I), multiply these components together to obtain the forecast value:

The forecasted value of Y is

$$\hat{Y} = T \times S \times C \times I = T \times \frac{S(\%)}{100} \times \frac{C(\%)}{100} \times \frac{I(\%)}{100}$$

2) If only two components are known, T and S , this approach is often preferred because T and S are easier to estimate and follow definite patterns. In this case, $C \times I$ is assumed to be

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approximately equal to 1, since C and I are components that vary in magnitude and cyclical timing.

Therefore, for forecasting purposes, $C \times I \approx 1$, and:

The forecasted value of Y is

$$\hat{Y} = T \times S = T \times \frac{S(\%)}{100}$$

Where: T = Estimated trend value from the equation derived using the least squares method

S = **Estimated seasonal variation**, also called the seasonal index, which can be calculated **using one of two methods**:

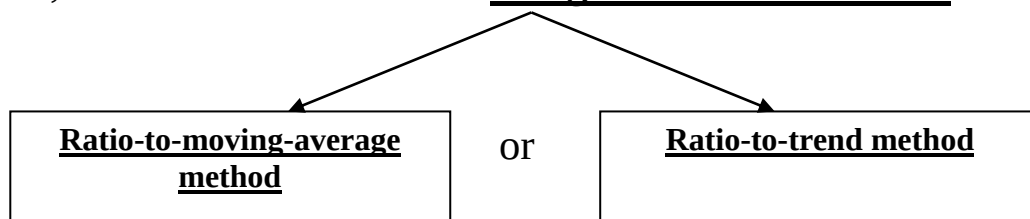


Figure: Comparison of Actual Sales Data (Y) versus Forecasted Sales Data $\hat{Y}$

Quarterly data from B.E. 2537 – 2544

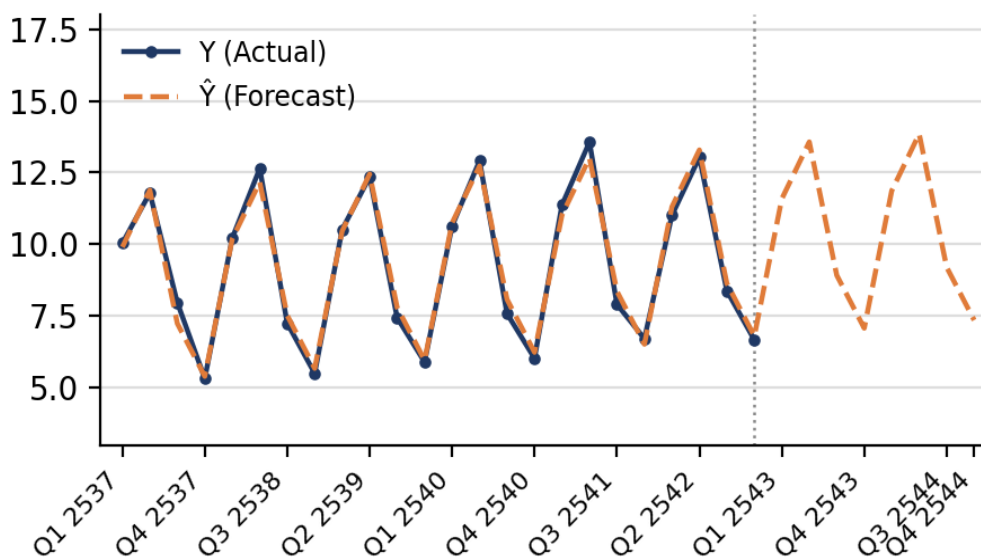


Table: Comparison of Actual Sales Data (Y) versus Forecasted**Sales Data $\hat{Y}$** **Quarterly data from B.E. 2537 – 2544**

Year	Quarter	Y (Million Baht)	X	$\hat{Y} = 8.11 + 0.09X$ (T)	S(%)	$\hat{Y} = T \times \frac{S(\%)}{100}$ (Let $C \times I = 1$)
2537	1	6.7	1	8.20	76.49	6.27
	2	4.6	2	8.29	57.47	4.77
	3	10.0	3	8.38	114.14	9.57
	4	12.7	4	8.47	151.90	12.87
2538	1	6.5	5	8.56	76.49	6.55
	2	4.6	6	8.65	57.47	4.97
	3	9.8	7	8.74	114.14	9.98
	4	13.6	8	8.83	151.90	13.41
2539	1	6.9	9	8.92	76.49	6.82
	2	5.0	10	9.01	57.47	5.18
	3	10.4	11	9.10	114.14	10.39
	4	14.1	12	9.19	151.90	13.96
2540	1	7.0	13	9.28	76.49	7.10
	2	5.5	14	9.37	57.47	5.39
2541	3	10.8	15	9.46	114.14	10.80
	4	15.0	16	9.55	151.90	14.51
	1	7.1	17	9.64	76.49	7.37
	2	5.7	18	9.73	57.47	5.59
	3	11.1	19	9.82	114.14	11.21
2542	4	14.5	20	9.91	151.90	15.05
	1	8.0	21	10.00	76.49	7.65
	2	6.2	22	10.09	57.47	5.80
	3	11.4	23	10.18	114.14	11.62
	4	14.9	24	10.27	151.90	15.60
2543	1		25	10.36	76.49	7.92
	2		26	10.45	57.47	6.01
	3		27	10.54	114.14	12.03
	4		28	10.63	151.90	16.14
2544	1		29	10.72	76.49	8.20

Year	Quarter	Y (Million Baht)	X	$\hat{Y} = 8.11 + 0.09X$ (T)	S(%)	$\hat{Y} = T \times \frac{S(\%)}{100}$ (Let $C \times I = 1$)
	2		30	10.81	57.47	6.21
	3		31	10.90	114.14	12.44
	4		32	10.99	151.90	16.69

Example: Using quarterly sales data, the trend equation, and seasonal indices, answer the following questions:

1) Estimate the **trend** values for sales in Quarters 1 through 4 of B.E. 2543

Given the quarterly sales data, the quarterly trend estimation equation is:

$$\hat{Y}_T = 8.11 + 0.09X$$

Where: $\hat{Y}_T$ = Estimated trend value of quarterly sales (in million baht)

X = Time period (in quarters)

Origin (X = 0) : November 15, B.E. 2536

Using this equation to estimate the trend values for sales in Quarters 1 through 4 of B.E. 2543

$$Q_{1/2543} : X = 25; \quad \hat{Y}_T = 8.11 + (0.09 \times 25) = 10.36 \text{ million baht}$$

$$Q_{2/2543} : X = 26 \quad \hat{Y}_T = 8.11 + (0.09 \times 26) = 10.45 \text{ million baht}$$

$$Q_{3/2543} : X = 27 \quad \hat{Y}_T = 8.11 + (0.09 \times 27) = 10.54 \text{ million baht}$$

$$Q_{4/2543} : X = 28 \quad \hat{Y}_T = 8.11 + (0.09 \times 28) = 10.63 \text{ million baht}$$

2) If sales in the first quarter of B.E. 2543 amounted to 9,000,000 baht and the cyclical variation is assumed to be constant ($C = 1$), to what extent does the irregular component affect the first quarter?

Find $I = ?$

Where $Y = 9,000,000 \text{ baht} = 9 \text{ million baht}$

$T = \hat{Y}_T$ = Forecasted trend value of sales for Quarter 1, B.E. 2543
= 10.36 million baht

$C = 1$

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$$S = \text{Seasonal index for Quarter 1} = \frac{76.49}{100}$$

From the model: $Y = T \times S \times C \times I$

$$\therefore I = \frac{Y}{T \times S \times C} = \frac{9}{10.36 \times \left(\frac{76.49}{100}\right) \times 1} = 1.1357$$

$$I(\%) = 1.1357 \times 100 = 113.57$$

In Quarter 1 of 2543, the irregular component had an influence, causing sales to be 13.57% **higher** than normal.

- 3) If sales in Quarter 2 of B.E. 2543 were 6,500,000 baht, determine the extent to which the cyclical and irregular components ($C \times I$) affected sales in Quarter 2.

Find $C \times I = ?$

From the model: $Y = T \times S \times C \times I$

$$C \times I = \frac{Y}{T \times S} = \frac{6.5}{10.45 \times \left(\frac{57.47}{100}\right)} = 1.0823$$

$$(C \times I)\% = 1.0823 \times 100 = 108.23$$

In Quarter 2 of B.E. 2543, the combined effect of the cyclical and irregular components ($C \times I$) increased sales above the normal level by 8.23%. ($108.23 - 100 = 8.23\%$)

- 4) Determine whether the seasonal effect causes sales in Quarter 4 to be higher or lower than normal, and by what percentage. (Compute the seasonal index for Quarter 4 and interpret the result. The seasonal index has already been calculated.)

Calculate the seasonal index for Quarter 4 and interpret its meaning

$S_{\text{ไตรมาส 4}}(\%) = 151.90$ (the seasonal index has already been computed)

This means that seasonal effects cause sales in Quarter 4 to be 51.90% **higher** than normal. ($151.90 - 100 = 51.90\%$)

- 5) If sales in Quarter 4 of B.E. 2543 were 15,000,000 baht, compute the deseasonalized (**seasonally adjusted**) sales, assuming **no seasonal influence**.

Calculate the deseasonalized data “Deseasonalized Data” (Deseasonalized Data) $\times 100$;

$$= \frac{Y}{S(\%)} \times 100$$

$$= \frac{15}{151.90} \times 100 = 9.87 \text{ million baht}$$

In Quarter 4 of B.E. 2543, after removing the seasonal effect, the estimated true sales are approximately **9.87 million baht**.

- 6) If the deseasonalized sales for Quarter 1 of B.E. 2543 are 12 million baht, determine the extent to which the cyclical component affects sales in Quarter 1 of B.E. 2543, assuming that there is no irregular variation in any quarter.

1. The formula for deseasonalized sales is:

$$\text{Deseasonalized Sales} = \frac{Y}{S} = T \times C \times I$$

2. Since there is no irregular variation ($I = 1$) and the deseasonalized sales equal 12 million baht, we can set up the equation:

$$T \times C = 12$$

3. From Question 1, the trend value (T) for Quarter 1 of B.E. 2543 is 10.36 million baht. Substituting this value into the equation gives:

$$10.36 \times C = 12$$

$$C = \frac{12}{10.36} \approx 1.1583$$

$$C(\%) = 1.1583 \times 100 = 115.83\%$$

In Quarter 1 of B.E. 2543, the cyclical component caused sales to be **15.83% higher** than normal ($115.83\% - 100\%$).

- 7) Forecast sales for Quarters 1–4 of B.E. 2546 using the model

$$Y = T \times S, \text{ assuming that } C \times I = 1$$

Table: Forecasted sales ($\hat{Y}$: million baht) for Quarters 1–4 of B.E. 2546

Quarter	X	$\hat{Y} = 8.11 + 0.09X$	$S(\%)$	$\hat{Y} = T \times S = T \times \frac{S(\%)}{100}$
1	37	$8.11 + 0.09(37) = 11.44$	76.49	$11.44 \times 0.7649 = 8.75$
2	38	$8.11 + 0.09(38) = 11.53$	57.47	$11.53 \times 0.5747 = 6.63$
3	39	$8.11 + 0.09(39) = 11.62$	114.14	$11.62 \times 1.1414 = 13.26$
4	40	$8.11 + 0.09(40) = 11.71$	151.90	$11.71 \times 1.5190 = 17.79$

8) Forecast sales for Quarter 2 of B.E. 2553, assuming that the **cyclical effect** increases sales by 5% **above normal**, and the **irregular effect** decreases sales by 80% **below normal**.

1. **Find the time period (X) for Quarter 2 of B.E. 2553:**

- B.E. 2553 is exactly 10 years after B.E. 2543 ($10 \times 4 = 40$ quarters).
- Since Quarter 2 of B.E. 2543 has $X = 26$, Quarter 2 of B.E. 2553 will have $X = 26 + 40 = 66$.

2. **Calculate the trend value (T):**

$$T = 8.11 + 0.09(66) = 14.05 \text{ million baht}$$

3. **Identify the component values from the problem statement:**

- Seasonal index for Quarter 2 (S) = $57.47\% = 0.5747$
- Cyclical effect (C) = $100\% + 5\% = 105\% = 1.05$
- Irregular effect (I) = $100\% - 80\% = 20\% = 0.20$

4. **Calculate the forecasted sales (Y):**

$$Y = T \times S \times C \times I$$

$$Y = 14.05 \times 0.5747 \times 1.05 \times 0.20 \approx 1.70 \text{ million baht}$$

The forecasted sales for Quarter 2 of B.E. 2553 are **1.70 million baht** (or 1,695,689 baht).